

A Compact Counterexample to a Relative Point-Finite Cellularity Question

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Status

Status: counterexample likely valid, human review needed. This packet answers negatively a topological question raised in Rondos [1]: whether point-finite open families meeting a compact subset A of a compact space K must have cardinality at most the relative cellularity $c(A, K)$.

Source Question

Rondos defines, for $A \subseteq K$, the relative cellularity $c(A, K)$ as the supremum of sizes of cellular families of open subsets of K meeting A , and defines $c_{\text{pf}}(A, K)$ similarly using point-finite open families. The source paragraph containing the question is on page 8:

Further, it is natural to ask for which subsets A of a general compact space K , some of the numbers $c(A, K)$, $c_{\text{pb}}(A, K)$ and $c_{\text{pf}}(A, K)$ coincide. While it is well-known that the equality $c(A) = c_{\text{pb}}(A)$ holds for each topological space A , the equality $c(A, K) = c_{\text{pb}}(A, K)$ for a general subset A of K seems to be unclear. It was proved in [45, Lemma 1.2] (see also [50, Lemma 2.2], [46, Lemma 4.2] or [49, Theorem 3.8] for slightly different proofs) using the Baire category theorem, that in a compact ccc space, point-finite families of open sets must be countable. Thus one might expect that if $\mathcal{U}$ is a point-finite family of open sets which is meeting a subset A of a compact space K , then $|\mathcal{U}| \leq c(A, K)$. While we do not know whether this holds or not, the direct modification of any of the above proofs to the relative setting and to the case of arbitrary large cardinalities, only seems to work when the set A is open, or, more generally, when A is contained in the closure of its interior. If this were true also when the set A is compact, most of the results of this paper for Banach spaces E that do not contain c_0 and those with nontrivial cotype would be united.

In particular, the source asks whether a point-finite open family $\mathcal{U}$ meeting A must satisfy $|\mathcal{U}| \leq c(A, K)$, and notes that the compact- A case would unite parts of the paper's Banach-space results.

Proof Intuition

Use a standard Ψ -space idea, but choose the almost disjoint family so that any two large basic neighbourhoods have one prescribed isolated point in common. Let the compact set A be the one-point compactification of the index set. For each index i , there is a basic open set U_i meeting A at i . The family $\{U_i\}$ is point-finite because each isolated intersection point belongs to exactly two such sets.

However, cellular families meeting A cannot be uncountable. If an uncountable cellular family existed, the finite sets of deleted intersection points would form an uncountable finite-set mapping with no free pair. The Delta-system lemma gives a free pair, a contradiction. A countable cellular family does exist, so the relative cellularity is exactly ω .

Counterexample

Theorem 1. *Let κ be any uncountable cardinal. There exist a compact Hausdorff space K , a compact subset $A \subseteq K$, and a point-finite family $\mathcal{U}$ of open subsets of K meeting A such that*

$$|\mathcal{U}| = \kappa \quad \text{but} \quad c(A, K) = \omega.$$

Consequently the inequality $c_{\text{pf}}(A, K) \leq c(A, K)$ fails even for compact A .

Proof. Let $D = [\kappa]^2$ be the set of two-element subsets of κ . Write $p_{ij} = p_{ji}$ for the point $\{i, j\} \in D$. For $i < \kappa$, put

$$D_i = \{p_{ij} : j < \kappa, j \neq i\}.$$

The family $\{D_i : i < \kappa\}$ is almost disjoint, since $D_i \cap D_j = \{p_{ij}\}$ when $i \neq j$.

Let Ψ be the corresponding locally compact Ψ -space. Its underlying set is

$$D \cup \kappa.$$

Every point of D is isolated, and a neighbourhood base at $i < \kappa$ is given by

$$N(i, F) = \{i\} \cup (D_i \setminus F), \quad F \in [D_i]^{<\omega}.$$

This is Hausdorff because distinct $i, j < \kappa$ can be separated by deleting the single point p_{ij} from one of the two basic neighbourhoods. It is locally compact because each $N(i, F)$ has compact closure contained in $\{i\} \cup D_i$, a one-point compactification of the discrete set $D_i \setminus F$.

Let $K = \Psi \cup \{\infty\}$ be the one-point compactification of Ψ , and put

$$A = \kappa \cup \{\infty\} \subseteq K.$$

The set A is compact. Indeed, every compact subset of Ψ meets κ in only finitely many points: if it met infinitely many indices, the open cover consisting of one basic neighbourhood for each such index, together with singletons for points of D , would have no finite subcover. Thus neighbourhoods of ∞ in K meet A in cofinite subsets of κ together with ∞ , while every $i < \kappa$ is isolated in the subspace A .

For each $i < \kappa$, define

$$U_i = \{i\} \cup D_i.$$

This is an open subset of K and $U_i \cap A = \{i\}$, so the family $\mathcal{U} = \{U_i : i < \kappa\}$ meets A . It is point-finite: the point $i \in A$ belongs only to U_i , the point p_{ij} belongs exactly to U_i and U_j , and ∞ belongs to none of the U_i . Hence $c_{\text{pf}}(A, K) \geq \kappa$.

It remains to compute $c(A, K)$. First, $c(A, K) \geq \omega$. Choose distinct indices $i_0, i_1, \dots$ and set

$$V_n = \{i_n\} \cup (D_{i_n} \setminus \{p_{i_m i_n} : m < n\}).$$

Each V_n is open and meets A , and the family $\{V_n : n \in \mathbb{N}\}$ is pairwise disjoint. Thus $c(A, K) \geq \omega$.

Conversely, let $\mathcal{V}$ be a cellular family of open subsets of K meeting A . At most one member of $\mathcal{V}$ contains ∞ ; if such a member exists, then all but finitely many points of κ lie in it, so only

finitely many other members of $\mathcal{V}$ can meet A . We may therefore discard this harmless member and assume that every member of $\mathcal{V}$ meets κ .

For each $V \in \mathcal{V}$, choose $i_V \in V \cap \kappa$. Since V is open at i_V , there is a finite set $F_V \subseteq \kappa \setminus \{i_V\}$ such that

$$\{i_V\} \cup (D_{i_V} \setminus \{p_{i_V j} : j \in F_V\}) \subseteq V.$$

If $V \neq W$, disjointness of V and W forces

$$i_W \in F_V \quad \text{or} \quad i_V \in F_W,$$

because otherwise the common point $p_{i_V i_W}$ would belong to both V and W .

We claim that the set $J = \{i_V : V \in \mathcal{V}\}$ is countable. If it were uncountable, apply the Delta-system lemma to the finite sets $F_V \cap J$. Passing to an uncountable subfamily, these finite sets form a Delta-system with root R . Choose $i_V \notin R$. Since the petals are pairwise disjoint, at most one other selected finite set contains i_V . Choose W in the subfamily so that $i_W \notin F_V$ and $i_V \notin F_W$. This contradicts the displayed alternative. Hence J is countable, and so every cellular family of open sets meeting A is countable.

Therefore $c(A, K) = \omega$, while $\mathcal{U}$ is a point-finite family of size κ meeting the compact set A . $\square$

Scope and Limitations

This disproves the relative topological expectation stated in the source paragraph, including the compact- A version. It does not by itself disprove the separate Banach-space question of whether the c_{pb} conclusion in the source's main theorem can be upgraded from nontrivial cotype to the assumption that E contains no copy of c_0 . Rather, it shows that the proposed topological route through $c_{\text{pf}}(A, K) \leq c(A, K)$ is unavailable.

Verification and Novelty Check

Local run indexes were searched for arXiv:2305.06770, the paper title, relative cellularity, point-finite families, $c_{\text{pf}}(A, K)$, $c(A, K)$, Ψ -spaces, and almost disjoint family variants. No prior packet for this target was found. Web searches on 2026-07-03 for the exact relative cellularity and point-finite phrases, and for Ψ -space variants, surfaced the original arXiv page but no existing answer to this question.

The script `code/check_finite_incidence.py` checks only the finite incidence pattern of the construction. The proof above is the actual verification.

Human Review Recommendation

Review the compactness of A inside the one-point compactification and the Delta-system argument proving $c(A, K) \leq \omega$. These are the only non-routine points.

References

- [1] J. Rondon, *On isomorphisms of $\mathcal{C}(K)$ spaces and cardinal invariants of derivatives of K* , arXiv:2305.06770, 2023.